

Nayan Shankar Gupta

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Professional Summary

Second-year Computer Science Engineering student at VIT Vellore with a strong interest in AI systems, backend engineering, and scalable application development. Experienced in building full-stack and mobile applications using FastAPI, React, Flutter, PostgreSQL, and asynchronous architectures. Aspiring Backend and AI Engineer passionate about scalable distributed systems, intelligent workflows, and production-grade software development.

Professional Experience

Software Development Intern (On-site) – Bharuwa Solutions Pvt. Ltd., Haridwar May 2026 – June 2026
Flutter, Dart, SQLite

- Worked as an on-site Software Development Intern in an Agile development environment, contributing to the design and development of a cross-platform attendance management application.
- Collaborated with the development team to implement new features, debug issues, and improve application performance while following clean architecture principles.
- Gained hands-on experience with Flutter, SQLite, Git, and mobile application development throughout the complete software development lifecycle.

Creative Media Intern (On-site) – Arvasa Solution Pvt. Ltd. (Startup) July 2025 – October 2025
Canva, CapCut, Audacity

- Worked in a fast-paced startup environment to design promotional graphics and digital marketing creatives for branding and social media campaigns.
- Created and edited promotional videos using CapCut while enhancing audio quality with Audacity for marketing content.
- Collaborated with cross-functional teams to support branding initiatives, content planning, and campaign execution while meeting project deadlines.

Projects

AI-Orchestrator Platform *FastAPI, React.js, PostgreSQL, SQLAlchemy*

- Architected and developed a production-ready AI orchestration platform using FastAPI, React.js, and PostgreSQL, integrating **Gemini API** and **Groq API** through a unified multi-provider interface with intelligent provider fallback.
- Implemented secure JWT authentication, user preferences, persistent chat sessions, conversation history, and analytics dashboards, providing a personalized AI experience with 100% authenticated user access.
- Built an asynchronous orchestration engine supporting parallel API execution, response caching, and centralized provider management, reducing response latency by approximately **35–40%** compared to sequential requests.
- Designed scalable REST APIs using SQLAlchemy ORM, Alembic migrations, Pydantic, Dependency Injection, CORS, and Swagger (OpenAPI) documentation following modular backend architecture and clean software engineering practices.
- Developed a responsive React frontend featuring provider selection, session management, analytics visualization, and configurable AI preferences while establishing the foundation for future MCP, LangGraph, and AI agent workflow integration.

Flutter Attendance Management App *Flutter, Dart, SQLite*

- Developed a cross-platform attendance management application using Flutter and SQLite featuring secure authentication, offline data persistence, and responsive UI design.
- Implemented attendance tracking, punch in/out, work-hour calculation, leave management, calendar visualization, and attendance status classification (Present, Late, Half-Day, Absent).
- Integrated SharedPreferences, reusable UI components, and efficient local database management while following clean architecture and Git-based development workflows.

Education

Vellore Institute of Technology (VIT), Vellore 2024 – 2028
B.Tech in Computer Science Engineering (Information Security) CGPA: 8.97

Skills

Languages: Python, Java, C++, C, JavaScript, Dart, SQL
Backend: FastAPI, REST APIs, JWT Authentication, SQLAlchemy ORM, Alembic, Pydantic, Dependency Injection, Async Programming, CORS, Swagger (OpenAPI)
Frontend & Mobile: React.js, Flutter, Tailwind CSS, HTML5, CSS3, Responsive Design
Databases: PostgreSQL, SQLite
Tools & Platforms: Git, GitHub, VS Code, Android Studio, Render, Vercel
Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks